

RFID-RC522 Sensor

It senses your card/keychain and then turns on either a green or red led.

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Components and supplies

- 1 Arduino Uno Rev3
- 1 5 mm LED: Green
- 1 5 mm LED: Red
- 1 RFID Module (Generic)
- 11 Jumper wires (generic)
- 1 RFID Module (Generic)
- 1 Breadboard (generic)

Project description

I am trying to make a lock for my door like the ones at hotels, and also the leds the green one goes to pin 7 and red goes to pin 6 on the big power side (longer side) and the other side (smaller side) goes to GND

Code

```
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 10
#define RST_PIN 9
#define RED 2
#define Green 3
MFRC522 mfr522(SS_PIN, RST_PIN); // Create MFRC522
instance.
```

```

void setup()
{
  Serial.begin(9600);  // Initiate a serial communication
  SPI.begin();        // Initiate SPI bus
  mfrc522.PCD_Init();  // Initiate MFRC522
  Serial.println("Approximate your card to the
reader...");
  Serial.println();
  pinMode(LED_BUILTIN, OUTPUT);
  digitalWrite(LED_BUILTIN, LOW);  // turn the LED off
by making the voltage LOW
  digitalWrite(Green, LOW);
  digitalWrite(RED, LOW);
}

void loop()
{
  // Look for new cards
  if ( ! mfrc522.PICC_IsNewCardPresent())
  {
    return;
  }
  // Select one of the cards
  if ( ! mfrc522.PICC_ReadCardSerial())
  {
    return;
  }
  //Show UID on serial monitor
  Serial.print("UID tag :");
  String content= "";
  byte letter;
  for (byte i = 0; i < mfrc522.uid.size; i++)
  {

```

```

        Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" :
" ");
        Serial.print(mfrc522.uid.uidByte[i], HEX);
        content.concat(String(mfrc522.uid.uidByte[i] < 0x10
? " 0" : " "));
        content.concat(String(mfrc522.uid.uidByte[i], HEX));
    }
    Serial.println();
    Serial.print("Message : ");
    content.toUpperCase();
    if ((content.substring(1) == "1A 7C 78 30") ||
(content.substring(1) == "CC CC A3 16") ) //change here
the UID of the card/cards that you want to give access
    {
        Serial.println("Authorized access");
        Serial.println();
        digitalWrite(LED_BUILTIN, HIGH);    // turn the LED on
(HIGH is the voltage level)
        digitalWrite(Green, HIGH);
        delay(10000);
        digitalWrite(Green, LOW);
    }

    else    {

        Serial.println(" Access denied");
        digitalWrite(LED_BUILTIN, LOW);    // turn the LED
off by making the voltage LOW
        digitalWrite(RED, HIGH);
        delay(10000);
        digitalWrite(RED, LOW);

    }
}
}

```

```

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#include <MFRC522.h>

#define SS_PIN 10
#define
    RST_PIN 9
#define RED 2
#define Green 3
MFRC522 mfrc522(SS_PIN, RST_PIN);
    // Create MFRC522 instance.

void setup()
{
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    // Initiate a serial communication
    SPI.begin();    // Initiate  SPI bus

    mfrc522.PCD_Init();    // Initiate MFRC522
    Serial.println("Approximate your
card to the reader...");
    Serial.println();
    pinMode(LED_BUILTIN, OUTPUT);

    digitalWrite(LED_BUILTIN, LOW);    // turn the LED off
by making the voltage
LOW
    digitalWrite(Green, LOW);
    digitalWrite(RED, LOW);
}

void
loop()
{
    // Look for new cards

```

```

if ( ! mfrc522.PICC_IsNewCardPresent())

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    return;
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// Select one of the cards
if ( ! mfrc522.PICC_ReadCardSerial())

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Serial.print("UID
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String content= "";
byte letter;
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    Serial.print(mfrc522.uid.uidByte[i] < 0x10
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    Serial.print(mfrc522.uid.uidByte[i], HEX);
    content.concat(String(mfrc522.uid.uidByte[i]
< 0x10 ? " 0" : " "));
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HEX));
}
Serial.println();
Serial.print("Message : ");
content.toUpperCase();

if ((content.substring(1) == "1A 7C 78 30") ||
(content.substring(1) == "CC

```

```

    CC A3 16") ) //change here the UID of the card/cards
that you want to give access

{
    Serial.println("Authorized access");
    Serial.println();
    digitalWrite(LED_BUILTIN,
HIGH);    // turn the LED on (HIGH is the voltage level)
    digitalWrite(Green,
HIGH);
    delay(10000);
    digitalWrite(Green, LOW);
}

else
{

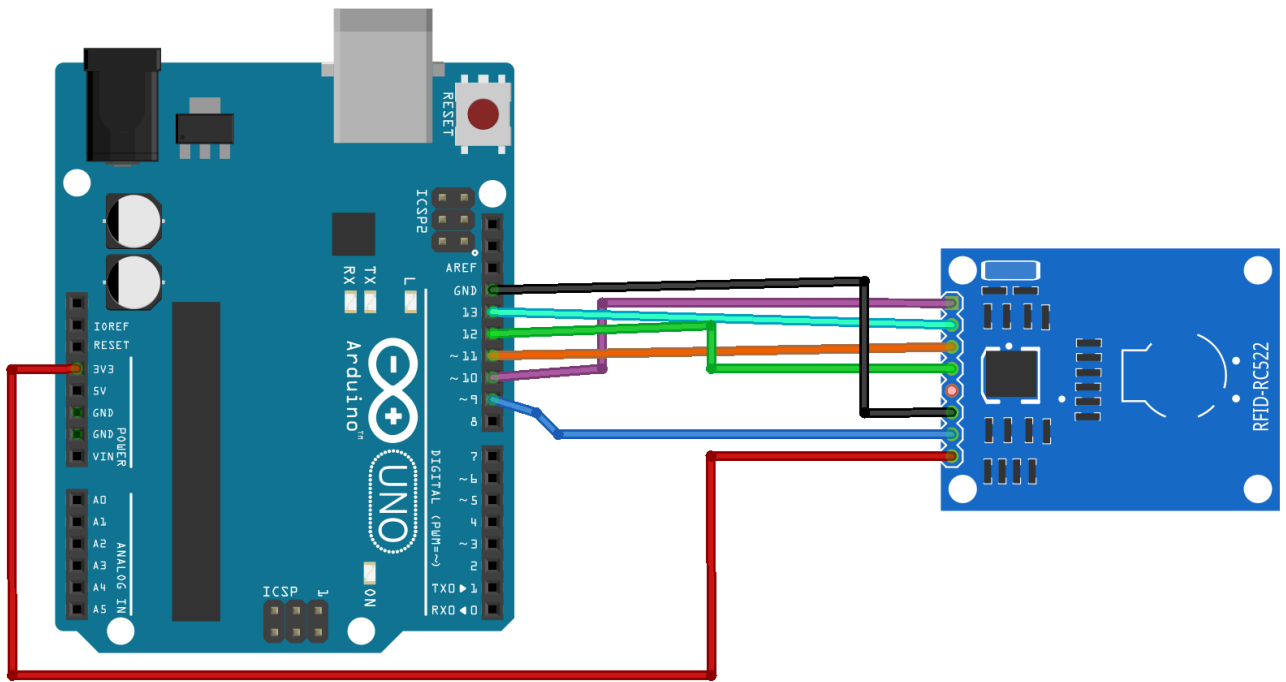
    Serial.println(" Access denied");
    digitalWrite(LED_BUILTIN,
LOW);    // turn the LED off by making the voltage LOW
    digitalWrite(RED,
HIGH);
    delay(10000);
    digitalWrite(RED, LOW);

}
}

```

Schematics

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Referencias

<https://projecthub.arduino.cc/person87/rfid-rc522-sensor-64000f>